

CORE GOALS OF CRYPTOGRAPHY

1. **Confidentiality** – the primary and most traditional goal of cryptography. It ensures that information remains accessible only to authorized users. It prevents unauthorized parties from reading or interpreting data by transforming it into unintelligible ciphertext.
2. **Integrity** – it guarantees that data has **not** been **altered**, modified, or tampered with **during transmission or storage**. If data is changed, cryptography allows the detection of this modification.
3. **Authentication** – **confirms** the identity of the sender, receiver, or device involved in a communication, ensuring that the parties involved are who they claim to be.
4. **Non-repudiation** – ensures that a sender cannot deny having sent a message or performed an action. It provides **legal and digital proof** of the origin of the data.

BASIC CRYPTOGRAPHIC TERMINOLOGIES

1. **Plaintext** (or cleartext) – the original, readable message or data before it is encrypted (e.g., text, images, or files).
*Basically, it acts as the **input** for encryption algorithms and the final **output** of decryption.*
NOTE: *Because it is encrypted, it is vulnerable to **interception** (unauthorized acquisition, tapping, or capturing of data while it is in transit between two or more communicating parties) and must be secured.*
2. **Ciphertext** – is encrypted, unreadable data produced by applying an encryption algorithm (cipher) to plaintext to ensure confidentiality. It appears as scrambled, random characters, numbers, or symbols that **cannot be understood without the correct description key**.
Basically, its primary purpose is to secure sensitive information from unauthorized access.
3. **Key** (cryptographic key) – a string of bits, numbers, or characters used by algorithms to lock (encrypt) or unlock (decrypt) data, transforming readable plaintext into secure ciphertext. These keys act like digital passwords, ensuring that only authorized parties with the correct key can access the information.
4. **Encryption** – the process of converting readable information (plaintext) into an unreadable format (ciphertext) using algorithms and secret keys to ensure confidentiality, integrity, and authentication. It prevents unauthorized parties from accessing sensitive data, both during transmission and in storage, by requiring a specific key for decryption.
5. **Decryption** – the authorized, mathematical process of converting unreadable ciphertext back into intelligible plaintext using a specific algorithm and key. It reverses encryption, ensuring only authorized users with the correct key can access protected data. It is essential for maintaining data confidentiality and integrity in secure communications.

*Basically, **Plaintext** is the readable input, **ciphertext** is the protected output, **keys** are the secret tools, **encryption** secures data, and **decryption** restores it for authorized use.*

CLASSICAL CRYPTOGRAPHIC ALGORITHMS

1. Caesar Cipher

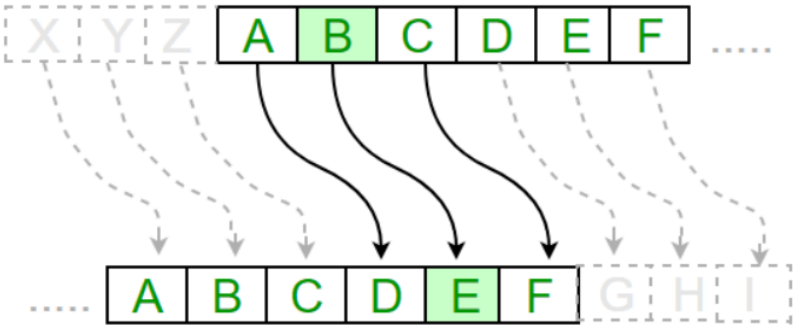
The Caesar cipher is one of the simplest and most widely known encryption techniques. It is a type of substitution cipher where *each letter in the plaintext is 'shifted' a certain number of places down the alphabet* (For example, with a shift of three, the letter 'A' becomes 'D', 'B' becomes 'E', and so on).

- It is named after Julius Caesar, who reportedly used it to protect his military communications.

Mathematical Formula:

$E_n(x) = (x + n)mod\ 26$
(Encryption Phase with shift n)

$D_n(x) = (x - n)mod\ 26$
(Decryption Phase with shift n)



EXAMPLES:

Text : ABCDEFGHIJKLMNOPQRSTUVWXYZ
Shift: 23
Cipher: XYZABCDEF GHIJKLMNOPQRSTUVW

Text : ATTACKATONCE
Shift: 4
Cipher: EXXEGOEXSRGI

ADVANTAGES	DISADVANTAGES
- Easy to implement and use thus, making suitable for beginners to learn about encryption. - Can be physically implemented, such as with a set of rotating disks or a set of cards, known as a scytale, which can be useful in certain situations. - Requires only a small set of pre-shared information. - Can be modified easily to create a more secure variant, such as by using a multiple shift values or keywords.	- It is not secure against modern decryption methods. - Vulnerable to known-plaintext attacks, where an attacker has access to both the encrypted and unencrypted versions of the same messages. - The small number of possible keys means that an attacker can easily try all possible keys until the correct one is found, making it vulnerable to a brute force attack. - It is not suitable for long text encryption as it would be easy to crack. - It is not suitable for secure communication as it is easily broken. - Does not provide confidentiality, integrity, and authenticity in a message.

Rules for the Caesar Cipher:

- Choose a number between 1 and 25. This will be your "shift" value.
- Write down the letters of the alphabet in order, from A to Z.
- Shift each letter of the alphabet by the "shift" value.
For example, if the shift value is 3, A would become D, B would become E, C would become F, and so on.
- Encrypt your message by replacing each letter with the corresponding shifted letter.
- To decrypt the message, simply reverse the process by shifting each letter back by the same amount.

REMEMBER CAESAR CIPHER'S KEY CHARACTERISTICS:

- Type: Substitution cipher
- Key: Single integer shift
- Alphabet size: 26

2. Monoalphabetic Substitution Cipher

This cipher improves upon the Caesar cipher by removing the fixed shift. Instead, any letter can be mapped to any other unique letter in the alphabet. It converts plain text into cipher text and re-convert a cipher text to plain text. Monoalphabetic Cipher eliminates the brute-force techniques for cryptanalysis. Moreover, the cipher line can be a permutation of the 26 alphabetic characters.

- This type of cipher is a form of **symmetric encryption** as the same key can be used to both **encrypt** and **decrypt** a message.

How Does Monoalphabetic Cipher Work?

Each letter of the plain text is replaced with another letter of the alphabet. It uses a fixed key which consist of the 26 letters of a “**shuffled alphabet**”.

Plain text alphabet:	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Cipher text alphabet (key):	M	U	A	L	V	O	Z	K	R	N	J	X	Q	D	F	S	H	P	E	B	C	T	I	W	Y	G

- One approach used to help decrypt a mono-alphabetic substitution cipher is to use a **frequency analysis** based on counting the number of occurrence of each letter to help identify the most recurrent letters. (e.g. In the English language, letters E, T and A).

Monoalphabetic Substitution Cipher

Plain text:

A long time ago, in a galaxy far, far away... It is a dark time for the Rebellion. Although the Death Star has been destroyed, Imperial troops have driven the Rebel forces from their hidden base and pursued them across the galaxy. Evading the dreaded Imperial Starfleet, a group of freedom fighters led by Luke Skywalker has established a new secret base on the remote ice world of Hoth. The evil lord Darth Vader, obsessed with finding young Skywalker, has dispatched thousands of remote probes into the far reaches of space...

1. Generate Key

Key: ZQEKRDXINFLTPMHAOWJBYSUCVVG

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

Z Q E K R D X I N F L T P M H A O W J B Y S C U V G

2. Start Substitution

Cipher text:

Z THMX BNPR ZXH, NM Z XZTZUV DZW, DZW ZCZV... NB NJ Z KZWL BNPR DHW BIR WRQRTTNHM. ZTBIHYXI BIR KRZBI JZBW IZJ QRRM KRJBWHVRK, NPARWNZT BWHHAJ IZSR KWNSRM BIR WRQRT DHWERJ DWHP BIRNW INKKRM QZJR ZMK AYWJYRK BIRP ZEWHJJ BIR XZTZUV. RSZKNMX BIR KWRZKRR NPARWNZT JBZWDTRRB, Z XWHYA HD DWRRKHP DNXIBRWJ TRK QV TYLR JLV CZTLRW IZJ RJBZQTNJIRK Z MRC JREWRB QZJR HM BIR WRPHBR NER CHWTK HD IHBI. BIR RSNT THWK KZWB I SZKRW, HQJRJJRK CNBI DNMKNMX VHVMX JLV CZTLRW, IZJ KNJAZBEIRK BIHYJZMKJ HD WRPHBR AWHQRJ NMBH BIR DZW WRZEIRJ HD JAZER...

ADVANTAGES	DISADVANTAGES
• Better Security than Caesar Cipher. • Provides Encryption and Decryption to data. • Monoalphabetic Cipher maintains a frequency of letters.	• Monoalphabetic ciphers are easy to break because they reflect the frequency data of the original alphabet. • Prone to guessing attack using the English letters frequency of occurrence of letters. • The English Language is used so the nature of plain text is known. • Less secure than a polyalphabetic cipher.

3. Vigenère cipher

The Vigenère cipher was designed to mask the letter frequencies that make monoalphabetic ciphers so vulnerable. It uses a series of interwoven Caesar ciphers. It uses a simple form of **polyalphabetic substitution**. A polyalphabetic cipher is any cipher based on substitution, using multiple substitution alphabets. The encryption of the original text is done using the **Vigenère square** or **Vigenère table**.

- It uses a **keyword** to determine the shift for each letter.
- The table consists of the alphabets written out 26 times in different rows, each alphabet shifted cyclically to the left compared to the previous alphabet, corresponding to the 26 possible Caesar Ciphers.
- At different points in the encryption process, the cipher **uses a different alphabet** from one of the rows.
- The alphabet used at each point depends on a repeating keyword.

Mathematical Representation: Where P is plaintext, C is ciphertext, and K is the repeating keyword:

$$C_i = (P_i + K_i) \pmod{26}$$

$$P_i = (C_i - K_i + 26) \pmod{26}$$

EXAMPLE:

Input : **Plaintext** : GEEKSFORGEEKS
Keyword : AYUSH

Output : **Ciphertext** : GCYCZFMLYLEIM

For generating key, the given keyword is repeated in a circular manner until it matches the length of the plain text. The keyword "AYUSH" generates the key "AYUSHAYUSHAYU"

The plain text is then encrypted using the process explained below.

Encryption:

The first letter of the plaintext, G is paired with A, the first letter of the key. So use row G and column A of the Vigenère square, namely G. Similarly, for the second letter of the plaintext, the second letter of the key is used, the letter at row E, and column Y is C. The rest of the plaintext is enciphered in a similar fashion.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

Table to encrypt Geeks

Decryption:

Decryption is performed by going to the row in the table corresponding to the key, finding the position of the ciphertext letter in this row, and then using the column's label as the plaintext. For example, in row A (from AYUSH), the ciphertext G appears in column G, which is the first plaintext letter. Next, we go to row Y (from AYUSH), locate the ciphertext C which is found in column E, thus E is the second plaintext letter.

ADVANTAGES	DISADVANTAGES
- Stronger than monoalphabetic substitution - Uses multiple alphabets - It flattens the frequency distribution of the ciphertext. A single plaintext letter (like 'E') will be encrypted as several different ciphertext letters depending on its position relative to the keyword, effectively thwarting simple frequency analysis.	- Vulnerable to: - Kasiski Examination - Frequency Analysis - Once the key length is discovered, the cipher becomes easier to break.

BASIC CRYPTANALYSIS TECHNIQUES

What is Cryptanalysis?

Cryptanalysis is the study of methods used to **break encryption systems** by discovering the plaintext, key, or algorithm without having the secret key.

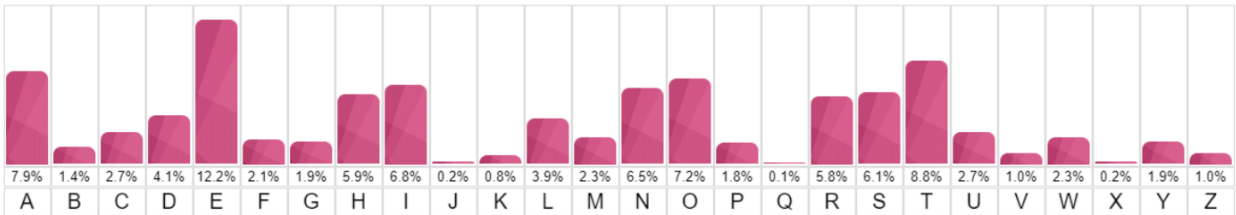
- The goal of cryptanalysis is to:**
 - Recover the original plaintext
 - Discover the secret key
 - Exploit weaknesses in the encryption method

In classical cryptography, two of the most important attack methods are:

1. Frequency Analysis

- In cryptography, frequency analysis is the study of the **frequency of letters or groups of letters** in a ciphertext. The method is used as an aid to breaking **substitution ciphers** (e.g. [mono-alphabetic substitution cipher](#), [Caesar shift cipher](#), [Vatsyayana cipher](#)).
- Frequency analysis consists of counting the occurrence of each letter in a text.
 - Frequency analysis is based on the fact that, in any given piece of text, certain letters and combinations of letters occur with varying frequencies. For instance, given a section of English language, letters E, T, A and O are the most common, while letters Z, Q and X are not as frequently used.

The following chart shows the frequency of each letter of the alphabet for the English language:



How the Attack Works:

Steps in frequency analysis:

- Obtain the ciphertext.
- Count how often each letter appears.
- Compare the frequencies with typical language statistics.
- Guess which ciphertext letters correspond to common plaintext letters.
- Gradually reconstruct the plaintext.

Example ciphertext: XLI UYMG0 FVSAR JSB NYQTW SZIV XLI PEDC HSK

The attacker:

- Counts letter frequencies
- Identifies common patterns
- Tests possible substitutions

Eventually the attacker may discover the plaintext.

Why Frequency Analysis Works:

It works against substitution ciphers because:

- Each plaintext letter always maps to the **same ciphertext letter**
- Statistical patterns remain visible

Ciphers Vulnerable to Frequency Analysis:

- *Frequency analysis easily breaks:*
 - Caesar Cipher
 - Monoalphabetic Substitution Cipher
- *It is less effective against:*
 - Vigenère Cipher (*because multiple alphabets are used. However, even Vigenère can eventually be broken.*)

2. Known-plaintext Attacks

- A known-plaintext attack occurs when the attacker has access to:
 - A piece of **plaintext**
 - The corresponding **ciphertext**
- Using this information, the attacker attempts to **discover the encryption key or algorithm**.

Principle:

If the attacker knows:

Plaintext: HELLO

Ciphertext: RIJVS

They can analyze the relationship between the two to determine the encryption method or key.

Example:

Suppose the encryption uses the Vigenère Cipher.

Plaintext: ATTACKATDAWN

Ciphertext: LXFOPEFRNHR

If the attacker knows both messages, they can compute:

$$\text{Key} = \text{Ciphertext} - \text{Plaintext}$$

Eventually revealing the keyword used in encryption.

Why do classical ciphers fail against modern attacks?

Fundamentally, classical ciphers fail because they were designed for an era of pencil-and-paper calculations. They lack the mathematical rigor required to withstand modern computational power and advanced statistical analysis.

- Classical ciphers were designed in an era when:

- Messages were short
- Attackers had limited computational power
- Cryptanalysis was manual

Modern computing makes them extremely insecure.

Primary reasons they fail:

1. Small Key Space

- Many classical ciphers have **very few possible keys**.
- Example:
 - The **Caesar Cipher** has only 25 possible keys
- *A computer can test all keys in milliseconds using **brute force**.*

2. Predictable Language Patterns

- Human languages contain:
 - predictable letter frequencies
 - common digraphs (TH, HE)
 - common trigraphs (THE, ING)
- *Because classical ciphers **do not hide these patterns**, they are easily analyzed using **Frequency Analysis**.*

3. Single Alphabet Substitution

- Many classical ciphers use **only one substitution alphabet**.
- Example:
 - **Plain:** ABCDEFG
 - **Cipher:** QWERTYU
- Because the mapping never changes:
 - $E \rightarrow Q$
 - $E \rightarrow Q$
 - $E \rightarrow Q$
- **Statistical patterns remain visible.**

4. Repeating Key Patterns

- Even stronger classical ciphers like the **Vigenère Cipher** repeat their key.
- Example:
 - **Key:** SECRETSECRETSECRET
- This repetition creates patterns that can be detected using techniques like the **Kasiski Examination**, which determines the key length.
- **Once the key length is known, the cipher becomes multiple simple substitution ciphers.**

5. Lack of Confusion and Diffusion

- Modern cryptography follows principles described by **Claude Shannon**:
 - **Confusion** – hide relationship between key and ciphertext
 - **Diffusion** – spread plaintext influence across ciphertext
- Classical ciphers lack these properties.
- **As a result:**
 - **Patterns remain visible**
 - **Keys are easier to deduce**

6. Computational Power

- Modern computers **can**:
 - test millions of keys per second
 - perform automated statistical analysis
 - run large-scale brute-force attacks
- **Therefore classical ciphers that were once secure are now trivial to break.**